

Lecture 7: OSI Model

*Address allocation follows a common pattern, whether for mobile phone numbers or IP addresses, and once devices have addresses, they still need a shared framework to actually communicate — this is what the **OSI Model** provides.*

How Phone Numbers Are Allocated

Mobile network operators (Idea, Airtel, Uninor, etc.) require phone numbers for their SIM cards, and the Indian Government allocates blocks of numbers to each of them — this works exactly like IP address allocation.

Phone number = 10 digits (0-9 range)

Idea → starts with 99_____ → 10^8 possible numbers

Airtel → starts with 86_____ → 10^8 possible numbers

Uninor → starts with 932_____ → 10^7 possible numbers

Tata Docomo → starts with 934_____ → 10^7 possible numbers

- ♦ Companies needing fewer SIMs (like Uninor) are given a longer fixed prefix, leaving fewer free digits and so a smaller block.
- ♦ Companies needing more numbers get a shorter fixed prefix, leaving more free digits and so a bigger block.

Why the country code (+91)?

*The same 10-digit number can exist in different countries. The **+91** country code tells the network which country's number is being dialed — this disambiguates numbers globally, similar to how a Network ID disambiguates devices.*

10^{10} = 1000 crore possible phone numbers in India
India's population $\approx$ 150 crore

Extended Subnetting Example

Take Idea's Class C block: 198.67.28.0

```
[ 198.67.28 | 0 ]  
  N/w ID      Host ID
```

$2^8 = 256$ total host addresses in this block

Dividing into 4 regional parts (Delhi, UP, UK, Haryana):

$256 \div 4 = 64$ addresses per region

REGION	BIT PATTERN (FIRST 2 BITS OF HOST ID)	RANGE SIZE
Delhi	00	64
UP	01	64
UK	10	64
Haryana	11	64

Delhi block: 00 000000 → 198.67.28.0
 00 111111 → 198.67.28.63

198.67.28.0/26 – Network ID uses 26 bits (24 original + 2 reserved for subnetting)
Host ID uses the remaining $32 - 26 = 6$ bits → $2^6 = 64$ addresses per subnet

UP block: 01 000000 → 198.67.28.64
 01 111111 → 198.67.28.127

General Rule for Dividing into Parts

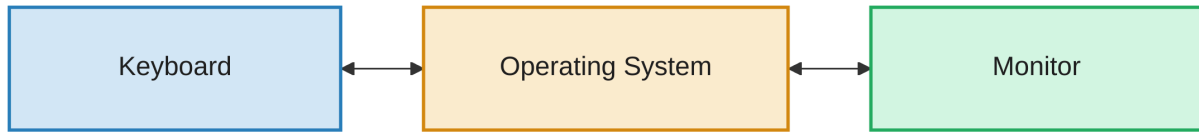
Half distribution (2 parts) → 1 bit reserved
4 parts → 2 bits reserved
8 parts → 3 bits reserved

*A block can only ever be divided into **powers of 2** — dividing into an odd number of parts is not possible, since each split simply uses one more bit.*

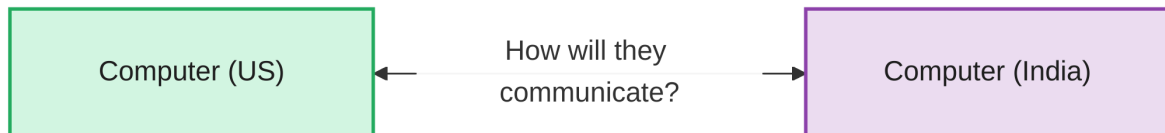
OSI Model — Open System Interconnection

Why Is a Common Model Needed?

Within a single computer, the **Operating System** already lets components like a monitor and keyboard communicate and interact seamlessly.



But now consider two separate computers — one in the US, one in India — that need to talk to each other:



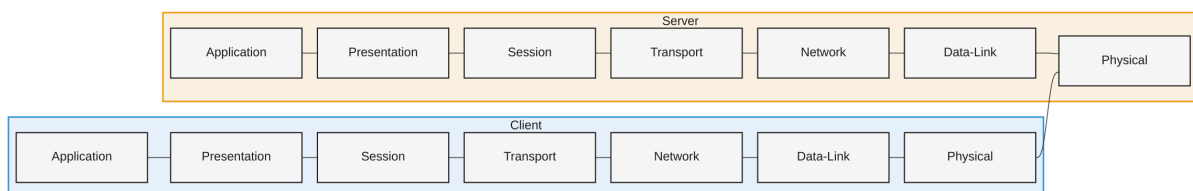
This raises real questions:

- ♦ How is data actually transferred between them?
- ♦ What protocols will be followed?
- ♦ How exactly will an HP machine and a Dell machine understand each other?

*The **OSI Model (Open System Interconnection)** is the standardized framework that answers these questions — a common set of layers that every system agrees to follow, so that any two devices, regardless of manufacturer, can communicate correctly.*

The 7 Layers of OSI

Application
Presentation
Session
Transport
Network
Data-Link
Physical



Data moves layer by layer — down from Application to Physical on the sending side, across the medium, then back up to Application on the receiving side.

1. Physical Layer

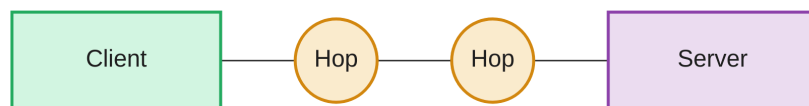
Handles how data actually travels at the most fundamental level — as electrical, optical, or radio signals over physical media.

Responsibilities

1. **Cables & Connectors** — the physical wiring that carries signals.
2. **Repeaters** — boost a signal's strength when it weakens over distance, so it can travel further without losing quality.
3. **Data Rate Control** — keeps the speed of data transfer under control (e.g. 3 Mbps, 10 Mbps).
4. **Encoding** — converting a digital signal into an analog signal and back.
5. **Physical Topology** — handling the physical layout being used, such as Bus, Ring, or Star.

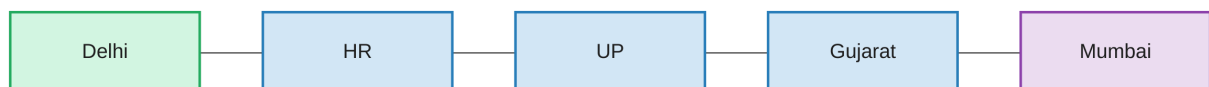
2. Data Link Layer

*Responsible for **Hop-to-Hop Delivery** — moving data reliably across one link at a time, from one intermediate device to the next.*



Hop-to-Hop refers to the intermediate path data takes — each hop is one segment of the total journey.

Analogy: Post Office



Hop-to-Hop delivery (e.g. Delhi → HR) is the responsibility of the Data Link Layer at each step along the way.

Layer Count: Router vs Server

DEVICE	LAYERS IT OPERATES AT
Router	3 layers — Network, Data Link, Physical
Device / Server	All 7 layers

Giving Physical Address — MAC Address

*The Data Link Layer is responsible for assigning the physical address, known as the **MAC Address** (Media/Ethernet Address).*

- ♦ **Windows:** `getmac /v /fo list`
- ♦ **macOS:** `networksetup -listallhardwareports`
- ♦ One device can have multiple MAC addresses — one for each way it can connect to a network, via its **NIC (Network Interface Card)**.

Number of ways to connect to the internet = Number of MAC addresses

Example: WiFi + Cable + Bluetooth = 3 ways → 3 different MAC addresses

Each hop carries a message in this structure:

[SIP | Data | RIP]

Having the correct address at each hop is critical — otherwise, data could be misdirected to the wrong destination.

- ♦ Data is transferred using the MAC address at this layer.
- ♦ A MAC address is device-specific and is never allocated to any other device.

Summary

- ♦ Address allocation for phone numbers works just like IP addressing — carriers get number blocks sized to their needs, and the +91 country code disambiguates numbers globally, similar to a Network ID.
- ♦ Subnetting can split a network block into any number of equal regions, but only in powers of 2 — dividing into 2, 4, or 8 parts reserves 1, 2, or 3 bits respectively; odd splits aren't possible.
- ♦ The OSI Model (Open System Interconnection) is a standardized 7-layer framework — Application, Presentation, Session, Transport, Network, Data-Link, and Physical — that lets any two devices, regardless of manufacturer, communicate correctly.
- ♦ Data moves layer by layer — down from Application to Physical on the sending side, across the medium, then back up to Application on the receiving side.
- ♦ The Physical Layer manages cables and connectors, repeaters (signal boosting), data rate control, encoding (digital-to-analog via modem), and physical topology.
- ♦ The Data Link Layer manages Hop-to-Hop delivery and assigns the Physical/MAC Address — a device-specific, unique identifier tied to its Network Interface Card (NIC).
- ♦ A device can have multiple MAC addresses, one per way it connects to a network (WiFi, cable, Bluetooth, etc.), while a router typically operates using only its bottom 3 layers (Network, Data Link, Physical), unlike end devices and servers, which use all 7.